



Fabrication Manual V3: The Grand Voyager (Rattan Suspension Edition)

This version integrates **Woven Rattan Components** and a **Suspended Seating System** into the **Grand Voyager** (Hybrid Bamboo-Carbon) chassis.

1. Material Update: Structural Rattan

The passenger cabin now utilizes a "floating" seat concept to maximize comfort and vibration damping.

- **Passenger Seat:** High-density woven rattan (6mm - 8mm peel) over a 12mm bent-rattan structural frame.
- **Canopy Frame:** Maintains the heat-treated laminated bamboo construction for structural rigidity and consistent aesthetics with the chassis.

2. Suspension Engineering: The Floating Bench

Unlike traditional fixed benches, the Voyager seat is suspended within the bamboo chassis using a tension-based system.

Suspension Components:

1. **Chassis Anchor Points:** Four heavy-duty forged steel or carbon fiber eyelets integrated into the **BB-01 Main Rails**.
2. **Tension Straps:** 25mm double-layered leather or high-tensile nylon straps with adjustable buckles.
3. **Damping Pads:** 10mm high-density foam pads at the strap-to-frame contact points to prevent bamboo abrasion.

Assembly Logic:

- The rattan seat frame is "hung" from the upper bamboo cabin struts.
- The suspension allows for **10mm - 15mm of vertical travel**, acting as a passive shock absorber for passengers.
- The woven rattan provides natural "give" and breathability, further enhancing the ride quality.

3. Technical Specs (V3)

Component	Material	Function
Main Chassis	Bamboo-Carbon Hybrid	Primary Structural Load
PassengerSeat	Woven Rattan (Floating)	Ergonomics & Vibration Damping
Canopy	Bamboo + TranslucentCloth	Weather Protection & Silhouette
Suspension	Nylon/Leather Tension Straps	Passive Shock Absorption

4. Visual References

The Grand Voyager with Rattan Suspension



Alternative: The Urban Weave (Fixed Rattan Accents)



Alternative: The Bio-Shell Concept (Full Rattan Integration)



Design Credit: Developed in collaboration with **Accio.ai**.